IR Proximity Sensor : Touchless Doorbell Module

❖ Summery

Pandemic serves worst for all and affected livelihood in many ways. Social distancing is the one of the best methods to escape from COVID-19. but, we can't avoid some emergency visits to some homes. When we arrived at In front of a house, first we search the doorbell button/ calling bell button. And press the button. But in this special situation this doorbell button can cause the virus to spread. so we came with this touchless idea.

❖ Objectives

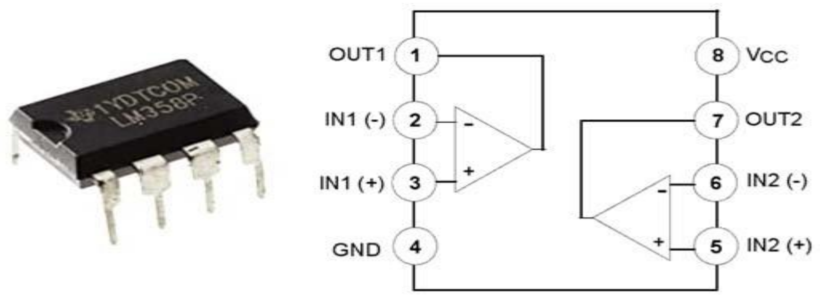
I. To learn the working of touchless doorbell.
II. To learn how to build simple circuit.
III. To learn working of IC358 as a OpAmp

❖ Components Required

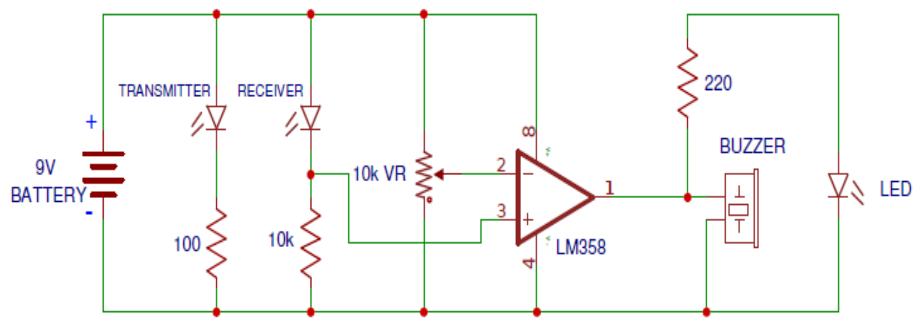
I. LM358 IC
II. Zero PCB Board
III. IR Transmitter Receiver Pair
IV. Buzzer
V. LED
VI. 10K ohm Variable Resistor
VII. Resistor (100 ohm, 220 ohm, 10K)
VIII. Some connecting wires

❖ LM358 IC as a OpAmp

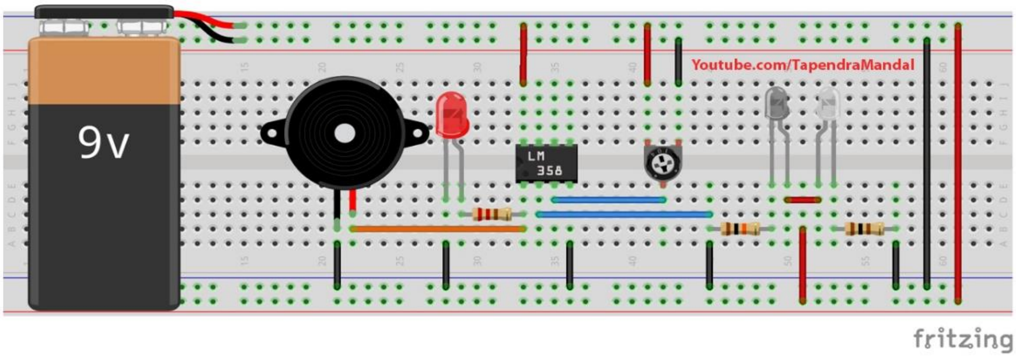
This sensor includes **IR transmitter LED and IR sensing LED. T**his sensor can be made by different ways but we are going to make it using LM358 comparator IC. This IC has two Operational Amplifier ports. We are going to use one side as we need single side sensing task. In each port of this IC three pins are available which are non - inverting terminal PIN 2, inverting terminal PIN 3 and an output PIN 1. When voltage at non- inverting terminal is higher than inverting terminal the output gets high which means it will provide a voltage output as signal.



❖ Circuit Diagram



❖ Breadboard Diagram



❖ Working

In above circuit we have connected pin 3 with photo diode which is non inverting terminal when IR rays will hit sensor LED the resistivity through it pins will be lowered and voltage supply will get high on the pin 3 which results as high output at PIN1. In our requirement we want to make a variable sensing device so that we can manipulate the sensitivity or set the value of intensity to be sensed. we add a variable resistor at pin 2. As the variable resistor is set for low resistance, the voltage on pin 2 gets high. The non- inverting terminal PIN 3 requires higher input signal to overcome the voltage on PIN 2 for higher output which means it is set for high intensity of IR light. As the resistance is improved the the sensitivity of the sensor increases due to lesser voltage requirement in PIN3.

❖ Application

I. Hand dryer
II. Door Bells
III. Automatic Doors

❖ Conclusion

In this IR Proximity sensor project, we learned working of IC358 IC as a operational amplifier, concepts of IR Transmitter and receiver, applications of this project and so many concepts we learned In this project.

Thank You!